

Exercise and worked solution

Options & strategies

Four parts to handle options : split a premium and read moneyness, apply put-call parity, build and price the strategies (spread, straddle), and reason through early exercise and exotics. Work through each part before reading the solution.

Part A · Intrinsic, time value and moneyness

PROBLEM

A stock trades at $S = 53$. A call with strike $K = 50$ trades at 4.5, a put of the same strike and maturity trades at 0.8. (a) Split each premium into intrinsic and time value. (b) What is each option's moneyness?

SOLUTION

- (a) Call: intrinsic = $\max(53 - 50, 0) = 3$, time value = $4.5 - 3 = \mathbf{1.5}$. Put: intrinsic = $\max(50 - 53, 0) = 0$, time value = $0.8 - 0 = \mathbf{0.8}$ (an OTM put is entirely time value).
- (b) The call is **ITM** ($S > K$), the put is **OTM**. A single strike/maturity pair cannot have both options ITM at once: one is ITM, the other OTM (or both ATM if $S = K$). At maturity time value falls to zero and the price reduces to the intrinsic.

Part B · Put-call parity

PROBLEM

A European call with strike $K = 50$, maturity 1 year, trades at $C = 5$. The stock is at $S = 52$, the annual risk-free rate is $R = 4\%$ (no dividend). By put-call parity, what is the price of the put with the same strike and maturity?

SOLUTION

- Parity: $C + PV(K) = P + S$, so $P = C + PV(K) - S$. $PV(K) = 50/1.04 = \mathbf{48.08}$.
- $P = 5 + 48.08 - 52 = \mathbf{1.08}$. The ITM call is logically worth more than the OTM put. Classic trap: forgetting to discount K and writing $P = 5 + 50 - 52 = 3$, which is wrong. Parity holds only in European style, for the same K and T .

Part C · Bull call spread and straddle

PROBLEM

(a) Bull call spread: long call $K = 48$ at 3.0, short call $K = 52$ at 1.2. Net cost, max gain, max loss, break-even? (b) ATM straddle: $S = K = 50$, call 2.12 and put 1.87. Cost, break-evens, profit at 58, max loss?

SOLUTION

- (a) Net cost = $3.0 - 1.2 = \mathbf{1.8}$. Max gain above 52: $(52 - 48) - 1.8 = \mathbf{2.2}$. Max loss below 48: $\mathbf{1.8}$ (both calls expire). Break-even = $48 + 1.8 = \mathbf{49.8}$. The profile is capped on both sides: one sells the upside beyond 52 to cut the outlay.
- (b) Cost = $2.12 + 1.87 = \mathbf{3.99}$. Payoff = $|S_t - 50| - 3.99$, break-evens at $50 \pm 3.99 = \mathbf{46.01 \text{ and } 53.99}$ ($\approx [46 ; 54]$). At 58: $8 - 3.99 = \mathbf{+4.01}$. Max loss = $\mathbf{3.99}$ when $S_t = 50$. It is the bet on movement, direction irrelevant.

Part D · Early exercise and exotics

PROBLEM

(a) Should one exercise an American call with strike 40, $S = 44$, on a non-dividend stock early, if its European lower bound is 4.49? And if the market shows it at 4.20? (b) Rank Asian, vanilla and lookback from cheapest to dearest, and state what one accepts with a knock-out barrier.

SOLUTION

- (a) Exercising yields only the intrinsic $44 - 40 = 4$; selling yields at least 4.49 (intrinsic + time value), so one **sells, never exercises** (without a dividend). At 4.20 the call trades below its bound of 4.49: this is an **arbitrage** (buy the call, short the stock, invest the cash). With a large dividend before maturity, exercising just before the ex-date can become optimal again.
- (b) **Asian < vanilla < lookback**: the average moves less than the final price (Asian cheaper), the extreme reached moves more (lookback dearer). A knock-out is cheaper but can disappear if the barrier is touched: one accepts being unhedged at the worst moment (gap risk). And knock-in + knock-out = vanilla.

Key takeaways: price = intrinsic + time value, moneyness ITM/ATM/OTM; put-call parity $C + PV(K) = P + S$ (European, discount K); bull spread capped at $K2 - K1$, straddle = bet on movement (leave the break-evens $K \pm$ total premium); never exercise an American call on a non-dividend stock early (selling gives the time value too); exotics order Asian < vanilla < lookback, and knock-in + knock-out = vanilla.